

TASK 4

CASE CLASSIFICATION & SEVERITY ANALYSIS

Classifying Adverse Drug Reactions by Seriousness and Expectedness

Assessment Methods

ICH E2A Seriousness Criteria + Expectedness Determination Against Reference Safety Information

Cases Reviewed

- Case 1 — Amoxicillin (Skin rash, itching)
- Case 2 — Ibuprofen (GI bleeding, hospitalization)
- Case 3 — Metformin (Lactic acidosis)
- Case 4 — Atorvastatin (Elevated liver enzymes, mild)
- Case 5 — Levofloxacin (Tendon rupture)

All cases in this document are fictional educational scenarios created for pharmacovigilance internship practice.

1. Learning Objectives

The purpose of this assignment is to demonstrate how a pharmacovigilance professional classifies an adverse drug reaction (ADR) along two independent dimensions: **seriousness** (the clinical impact and regulatory reporting priority of the event) and **expectedness** (whether the event is already documented in the reference safety information for the product).

These two dimensions are assessed separately. A reaction can be serious yet expected (e.g. a labeled reaction that happens to require hospitalization), or non-serious yet unexpected (e.g. a mild reaction not previously described for the drug). The combination of seriousness and expectedness determines regulatory reporting timelines, such as expedited 15-day reporting for serious and unexpected reactions.

2. Seriousness Criteria (ICH E2A)

An adverse event or reaction is classified as **serious** when it meets at least one of the following outcomes, regardless of the severity of the individual symptom:

Criterion	Definition
Death	The adverse event results in the patient's death.
Life-threatening	The patient was, in the view of the reporter, at immediate risk of death from the event as it occurred.
Hospitalization	The event requires inpatient hospitalization or prolongs an existing hospitalization.
Disability / Incapacity	The event results in persistent or significant disability or incapacity.

Congenital anomaly	The event is a congenital anomaly or birth defect in the offspring of an exposed patient.
Other medically important event	The event does not meet the criteria above but may jeopardize the patient or require intervention to prevent one of the outcomes listed, based on appropriate medical judgement.

A reaction is classified as **non-serious** when none of these criteria are met, even if the symptom is bothersome or clinically notable.

3. Expectedness Determination

An adverse reaction is **expected** when its nature, severity and specificity are consistent with the reference safety information for the product — typically the approved product label, package insert, or Company Core Data Sheet (CCDS). It is **unexpected** when it is not listed, or is listed but at a different severity or specificity than observed in the case.

Expectedness is a labeling-based, document-driven determination — not a judgement of how common or medically plausible the reaction is. A well-recognized class effect that is nonetheless absent from a specific product's approved labeling is still classified as unexpected for that product.

4. Evidence Checklist

Factor	Assessment Question	Why It Matters
Clinical outcome	Did the event lead to death, hospitalization, disability or was it life-threatening?	Directly determines seriousness classification.
Factor	Assessment Question	Why It Matters
Medical judgement	Even without meeting a listed criterion, could the event jeopardize the patient without intervention?	Captures 'other medically important' events.
Reference safety information	Is the reaction listed in the current label/CCDS?	Determines expectedness.
Severity/specificity match	If listed, does the observed severity or specificity match the label?	A listed reaction at unexpected severity is still unexpected.
Reporting timeline	Is the case both serious and unexpected?	Triggers expedited regulatory reporting (e.g. 15 calendar days).

CASE 1 — AMOXICILLIN

A. Case Summary

Suspect drug: Amoxicillin 500 mg orally **Adverse**

reaction: Skin rash and itching

Patient: 26-year-old female treated for a respiratory tract infection.

B. Narrative

Two days after starting amoxicillin, the patient develops a mild, localized skin rash on the forearms with associated itching. There is no swelling, breathing difficulty or systemic involvement. The patient continues the course under observation and the rash resolves with an over-the-counter antihistamine, without stopping the antibiotic or requiring medical intervention beyond symptomatic care.

C. Seriousness Assessment (ICH E2A Criteria)

Criterion	Met?	Case Evidence
Death	No	Patient fully recovered.
Life-threatening	No	No airway, breathing or circulatory compromise at any point.
Hospitalization	No	Managed entirely on an outpatient basis.
Disability/Incapacity	No	No lasting functional impairment.
Congenital anomaly	No	Not applicable to this case.
Other medically important	No	Self-limited, required only symptomatic antihistamine treatment.

Seriousness Classification: Non-serious

D. Expectedness Assessment

Skin rash and pruritus are well-recognized, commonly reported hypersensitivity reactions to amoxicillin and are described in the reference safety information for the product.

Expectedness Classification: Expected

E. Regulatory Reporting Implication

Combined classification: **Non-serious / Expected**. This combination follows routine (non-expedited) reporting timelines, as the event does not meet both the serious and unexpected criteria simultaneously.

F. Pharmacovigilance Learning Point

Mild cutaneous hypersensitivity reactions are common with beta-lactam antibiotics. Correctly classifying a case as non-serious/expected prevents unnecessary escalation, while the reaction should still be recorded to support ongoing signal monitoring of reaction frequency.

CASE 2 — IBUPROFEN

A. Case Summary

Suspect drug: Ibuprofen 400 mg orally, repeated dosing

Adverse reaction: Gastrointestinal bleeding requiring hospitalization

Patient: 61-year-old male, history of osteoarthritis, using ibuprofen regularly for pain relief.

B. Narrative

After approximately three weeks of regular ibuprofen use, the patient presents with black, tarry stools and light-headedness. He is admitted to hospital, where endoscopy confirms an upper gastrointestinal bleed. He receives blood transfusion and proton-pump inhibitor therapy, and ibuprofen is permanently discontinued. He is discharged after five days in stable condition.

C. Seriousness Assessment (ICH E2A Criteria)

Criterion	Met?	Case Evidence
Death	No	Patient recovered and was discharged.
Life-threatening	No	Hemodynamically stabilized without imminent risk of death.
Hospitalization	Yes	Admitted for 5 days for endoscopy, transfusion and monitoring.
Disability/Incapacity	No	No persistent functional impairment reported.
Congenital anomaly	No	Not applicable to this case.
Other medically important	Not assessed	Hospitalization criterion already satisfied.

Seriousness Classification: Serious

D. Expectedness Assessment

Gastrointestinal bleeding, including hemorrhage and ulceration, is a well-documented class effect of NSAIDs and is explicitly listed in the reference safety information for ibuprofen, including specific warnings for patients with risk factors such as older age and chronic use.

Expectedness Classification: Expected

E. Regulatory Reporting Implication

Combined classification: **Serious / Expected**. This combination follows routine (non-expedited) reporting timelines, as the event does not meet both the serious and unexpected criteria simultaneously.

F. Pharmacovigilance Learning Point

This case illustrates that a reaction can be serious while still being expected. Because GI bleeding is a known, labeled risk of NSAIDs, this case follows routine reporting timelines rather than expedited timelines, despite the hospitalization.

CASE 3 — METFORMIN

A. Case Summary

Suspect drug: Metformin 1000 mg orally, twice daily

Adverse reaction: Lactic acidosis

Patient: 68-year-old female with type 2 diabetes and recently declining renal function.

B. Narrative

The patient is brought to the emergency department with confusion, rapid breathing and abdominal pain. Laboratory testing reveals severely elevated lactate and metabolic acidosis. She is admitted to intensive care, metformin is discontinued, and she receives supportive management including fluids and bicarbonate therapy. Her condition is assessed by the treating team as having placed her at immediate risk of death before stabilization. She recovers after a prolonged intensive care admission.

C. Seriousness Assessment (ICH E2A Criteria)

Criterion	Met?	Case Evidence
Death	No	Patient survived following intensive care management.
Life-threatening	Yes	Treating team assessed immediate risk of death at presentation.
Hospitalization	Yes	Required ICU admission and prolonged inpatient stay.
Disability/Incapacity	Not assessed	Life-threatening criterion already satisfied.
Congenital anomaly	No	Not applicable to this case.
Other medically important	Not assessed	Life-threatening criterion already satisfied.

Seriousness Classification: Serious

D. Expectedness Assessment

While metformin-associated lactic acidosis is listed as a rare risk in the reference safety information, for the purposes of this training case it is treated as not clearly captured by the current label wording (e.g. differing in severity or population-specific risk factors from what is described), and is therefore assessed as unexpected. In real practice, the assessor must compare the exact wording of the current approved label against the observed event.

Expectedness Classification: Unexpected

E. Regulatory Reporting Implication

Combined classification: **Serious / Unexpected**. This combination meets the threshold for expedited regulatory reporting, as the event is both serious and unexpected.

F. Pharmacovigilance Learning Point

This is the highest-priority combination for regulatory purposes: serious and unexpected. It triggers expedited reporting timelines. This case also illustrates the importance of considering patient risk factors, such as declining renal function, when interpreting a known but rare labeled risk.

CASE 4 — ATORVASTATIN

A. Case Summary

Suspect drug: Atorvastatin 20 mg orally, once daily

Adverse reaction: Elevated liver enzymes (mild)

Patient: 52-year-old male started on statin therapy for hyperlipidemia.

B. Narrative

Routine liver function testing performed six weeks after starting atorvastatin shows a mild, asymptomatic elevation of liver transaminases, below three times the upper limit of normal. The patient has no jaundice, abdominal pain or other symptoms. Atorvastatin is continued at the same dose with a plan for repeat monitoring in four weeks.

C. Seriousness Assessment (ICH E2A Criteria)

Criterion	Met?	Case Evidence
Death	No	Not applicable.
Life-threatening	No	Asymptomatic finding on routine monitoring.
Hospitalization	No	Managed with outpatient monitoring only.
Disability/Incapacity	No	No functional impact.
Congenital anomaly	No	Not applicable to this case.
Other medically important	No	Mild, asymptomatic, did not require treatment interruption.

Seriousness Classification: Non-serious

D. Expectedness Assessment

Mild, asymptomatic transaminase elevation is a well-recognized, labeled effect of statin therapy, with routine monitoring specifically recommended in the reference safety information.

Expectedness Classification: Expected

E. Regulatory Reporting Implication

Combined classification: **Non-serious / Expected**. This combination follows routine (non-expedited) reporting timelines, as the event does not meet both the serious and unexpected criteria simultaneously.

F. Pharmacovigilance Learning Point

Routine monitoring often detects mild laboratory abnormalities that are already anticipated in a drug's safety profile.

Correctly classifying such findings as non-serious/expected avoids unnecessary alarm while the finding is still documented for ongoing safety monitoring.

CASE 5 — LEVOFLOXACIN

A. Case Summary

Suspect drug: Levofloxacin 500 mg orally, once daily

Adverse reaction: Tendon rupture

Patient: 57-year-old male, recreational runner, treated for a urinary tract infection.

B. Narrative

Approximately ten days into levofloxacin therapy, the patient experiences sudden sharp pain in the Achilles tendon while walking, followed by inability to bear weight normally on the affected leg. Clinical examination and imaging confirm a partial Achilles tendon rupture. Levofloxacin is discontinued immediately, the patient is referred for orthopedic assessment, and he is placed in a supportive boot with a structured rehabilitation plan. No inpatient admission is required, but the injury results in a functional limitation during the recovery period.

C. Seriousness Assessment (ICH E2A Criteria)

Criterion	Met?	Case Evidence
Death	No	Not applicable.
Life-threatening	No	No risk to life at any point.
Hospitalization	No	Managed on an outpatient orthopedic basis.
Disability/Incapacity	Yes	Partial tendon rupture caused a significant temporary functional limitation requiring structured rehabilitation.
Congenital anomaly	No	Not applicable to this case.
Other medically important	Not assessed	Disability criterion already satisfied.

Seriousness Classification: Serious

D. Expectedness Assessment

Tendon rupture, including Achilles tendon involvement, is a well-documented class effect of fluoroquinolone antibiotics and carries a boxed warning in the reference safety information for levofloxacin, with recognized risk factors including physical activity level and age over 60.

Expectedness Classification: Expected

E. Regulatory Reporting Implication

Combined classification: **Serious / Expected**. This combination follows routine (non-expedited) reporting timelines, as the event does not meet both the serious and unexpected criteria simultaneously.

F. Pharmacovigilance Learning Point

This case reinforces that seriousness and expectedness are assessed independently: a significant, disability-level injury can still be a fully expected, labeled risk. Because it is serious but expected, the case follows routine rather than expedited reporting timelines, though the causality and risk-factor context remain important to document.

5. Case Summary Table

The table below mirrors the portfolio tracker summary: five cases were reviewed, of which three were classified as serious and two as non-serious.

Case	Drug	Adverse Reaction	Seriousness	Expectedness
01	Amoxicillin	Skin rash, itching	Non-serious	Expected
02	Ibuprofen	GI bleeding, hospitalization	Serious	Expected
03	Metformin	Lactic acidosis	Serious	Unexpected
04	Atorvastatin	Elevated liver enzymes (mild)	Non-serious	Expected
05	Levofloxacin	Tendon rupture	Serious	Expected

Overall totals: Serious = 3 (Cases 2, 3, 5) | Non-serious = 2 (Cases 1, 4). **Expedited reporting priority:** Case 3 only (serious and unexpected).

6. Practical Pharmacovigilance Takeaways

1. **Assess seriousness and expectedness separately.** One does not determine the other; both must be documented with their own supporting rationale.
2. **Apply the ICH E2A criteria literally.** A single criterion met is sufficient to classify a case as serious, regardless of how the other criteria are assessed.
3. **Expectedness is a labeling exercise, not a clinical judgement.** Always compare against the current, version-controlled reference safety information rather than general medical knowledge of how common a reaction is.
4. **Identify the expedited reporting cases early.** Serious and unexpected cases carry the tightest regulatory timelines and should be flagged for priority processing.
5. **Document reasoning, not just the final label.** A reviewer should be able to see why each criterion was or was not met, not only the final classification.

7. Final Portfolio Conclusion

Across the five cases, seriousness ranged from non-serious outpatient findings to a life-threatening ICU admission, while expectedness varied independently of severity. Case 3 (metformin-associated lactic acidosis) represents the highest regulatory priority in this set, being both serious and unexpected. Cases 2 and 5 demonstrate that serious outcomes can still be fully expected, labeled risks, while Cases 1 and 4 illustrate routine, non-serious, expected findings that nonetheless warrant documentation for ongoing safety monitoring.

Educational Disclaimer: All cases are fictional training examples. They are not real patient records, real ICSRs or clinical advice.